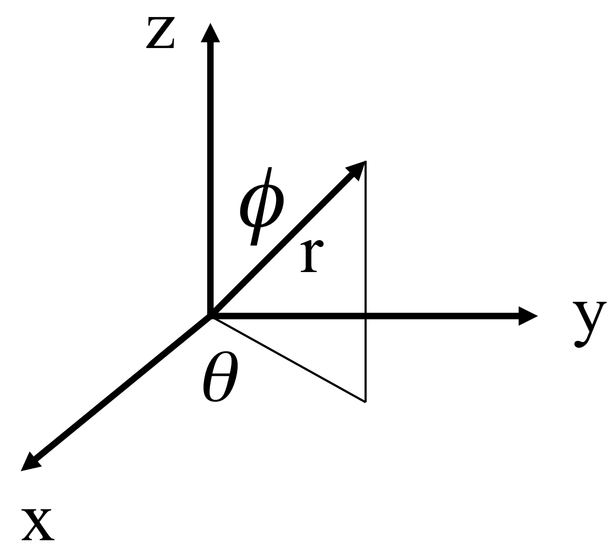
Internal Motion: Radial and Angular Motion

John W Daily

Internal Motion

It turns out to be convenient to work in spherical coordinates



$$\frac{\hbar^2}{2\mu}\nabla^2\psi_{\text{int}} + (\varepsilon_{\text{int}} - V(\vec{r}))\psi_{\text{int}} = 0$$

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\right]\psi_{i\,\text{int}}$$

$$+\frac{2\mu}{\hbar^2}\left[\varepsilon_{\text{int}} - V_{\text{int}}(r)\right]\psi_{\text{int}} = 0$$

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Again we pursue separation of variables

$$\psi_{\text{int}}(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2\mu r^2}{\hbar^2}\left[\varepsilon_{\text{int}} - V_{\text{int}}(r)\right] = -\frac{1}{Y}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]$$

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Calling the separation variable α, we get a radial equation and one that contains all the angular dependence

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left\{\frac{2\mu r^2}{\hbar^2}\left[\varepsilon_{\text{int}}-V_{\text{int}}(r)\right]-\alpha\right\}R=0$$

$$\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]=-\alpha Y$$

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Now separate the angular wave equation

$$Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$$

Obtaining

$$\frac{d^2\Phi}{d\phi^2} + \beta\Phi = 0$$

and

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) + \left(\alpha - \frac{\beta}{\sin^2\theta}\right)\Theta = 0$$

Where β is the separation constant. Note neither equation depends on V(r) so they can be solved once and for all.

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The general solution for Φ is simply (look up Euler's Formula)

$$\Phi(\phi) = \exp(i\beta^{1/2}\phi)$$

Note that this function must be continuous and single valued, so

$$\Phi(\phi) = \Phi(\phi + 2\pi)$$

This is so only if $\beta^{1/2} = m_l$, $m_l = 0, \pm 1, \pm 2$ etc

Thus

$$\Phi(\phi) = \exp(i m_l \phi)$$

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Plugging the expression for β in the equation for Θ

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right)+\left(\alpha-\frac{m_l^2}{\sin^2\theta}\right)\Theta=0$$

This is a Legendre Equation. To satisfy the boundary conditions we must have

$$\alpha=l(l+1)\,,\qquad l=i+|m_l|,\quad i=0,1,2,3,\ldots$$

$$\Theta=P_l^{|m_l|}(\cos\theta)=\frac{1}{2^l\,l!}(1-\cos^2\theta)^{|m_l|/2}\frac{d^{|m_l|+l}}{d(\sin\theta)^{|m_l|+l}}(\sin^2\theta-1)^l$$

associated Legendre function

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Thus the angular component of the wave function becomes

$$Y(\theta,\phi) = C_{l,m_l} P_l^{|m_l|}(\cos\theta) e^{im_l\phi}$$

Where C is the normalization constant

$$C_{l,m_l} = \frac{1}{(2\pi)^{1/2}} \left[\frac{(2l+1)(l-|m_l|)!}{2(l+|m_l|)!} \right]^{1/2}$$

Significance of l and m_l

Recall that

$$\langle B \rangle = \int \Psi^*(\vec{r},t) B_{op} \Psi(\vec{r},t) dV$$

If we calculated the angular momentum of our system we would get

$$\langle L^2 \rangle = l(l+1)\hbar^2$$

Thus l determines the expectation value of the angular momentum and is called the angular momentum quantum number

Significance of l and m_l

If we calculated the z component of angular momentum

$$\langle L_z \rangle = m_l \hbar$$

Normally, atoms or molecules are randomly oriented. However, in the presence of a magnetic field they will align. Thus m_l is called the magnetic quantum number